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Elemental Sulfur Management in Acid Gas Field Development: A Case Study from Uzbekistan

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Abstract

This paper examines development challenges related to elemental sulfur content in reservoir fluid from an acid gas field in the Afghan-Tajik Basin of Uzbekistan. The productive reservoirs, located within Jurassic subsalt carbonate formations, contain natural gas saturated with gaseous sulfur.

Our study evaluates the impact of sulfur condensation and solidification on well and reservoir performance and presents an integrated approach for incorporating these effects into field development planning.

We developed an empirical model of sulfur deposition within well tubing that accounts for multiphase flow regimes and pressure-temperature profiles along the wellbore. The model builds upon laboratory experiments on sulfur solubility in sour gases (Brunner et al. 1980, 1988) and utilizes sulfur phase diagram. Field-observed solid deposition rates on tubing walls were used to calibrate the model.

To investigate reservoir effects, we incorporated sulfur solubility and relative permeability functions for a three-phase reservoir system (gas – liquid sulfur – water) into a reservoir simulation model. Local grid refinements enabled simulation of liquid sulfur banking in near-wellbore zones (NWZ) and analysis of this phenomenon's impact on well productivity.

Our research identifies two primary consequences of sulfur phase transforms on production well performance:

- Reduced well productivity due to sulfur banking in near-wellbore zone
- Restricted tubing flow from solid sulfur deposition

Simulation results indicate that productivity index reduction due to sulfur banking is dependent on pressure drawdown and may reach 30% during the first year of production – a significant effect that must be addressed in field development design. Solid sulfur deposition on tubing walls predominantly occurs during annular multiphase flow regime. Progressive deposition gradually constricts flow, potentially resulting in complete blockage within several days to a month, depending on gas rate and pressure-temperature profiles. Notably, deposition progresses sufficiently slower at mist flow regime. Therefore, to maximize

tubing efficiency, either mist multiphase flow or annular flow with minimal pressure/temperature drawdown should be maintained. Both consequences of sulfur phase transforms - reduced well productivity and restricted tubing flow – significantly affect field production and are sensitive to well operational parameters. By integrating our empirical tubing deposition model with reservoir simulation model, we developed a multiparameter optimization framework that minimizes the combined negative effect of sulfur phase transforms on field production. This approach enables selection of optimal well count, well construction specifications, well operational constraints, and solvent treatment schedule.

The substantial negative impact of sulfur condensation and solidification on well productivity and operational efficiency represents a critical challenge for acid gas field development. This paper presents a practical approach implemented in an Uzbekistan field that addresses sulfur phase behavior in both the reservoir and wellbore tubing through the integration of reservoir simulation and empirical deposition modeling.

Introduction

The development of natural gas fields with high hydrogen sulfide content is complicated by the presence of dissolved elemental sulfur in the gas. While the mechanism of elemental sulfur formation may vary depending on geochemical conditions, laboratory studies have confirmed that elemental sulfur is frequently present in stable equilibrium within hydrogen sulfide gas. This phenomenon has been systematically characterized by laboratory studies determining elemental sulfur solubility in sour gases (Brunner et al. 1980, 1988). These comprehensive studies laid the foundation for quantitative assessment of equilibrium sulfur concentrations in the gas phase under various pressure and temperature conditions.

A decrease in pressure, temperature, or change in gas composition during production reduces the equilibrium solubility of sulfur, causing the gas to exceed the solubility limit and precipitate elemental sulfur as liquid or solid phase. When not entrained by the gas flow, the deposited sulfur accumulates in the reservoir, near-wellbore zone, on tubing walls, and in surface facilities. This accumulation reduces reservoir gas phase permeability, decreases effective tubing cross-sectional area, increases frictional pressure losses, and causes operational interruptions. Consequently, these effects lead to reduced well productivity and diminished economic efficiency of field development.

The issue of elemental sulfur precipitation during the operation of sour gas wells has been recognized since the mid-20th century. In the work of Kuo and Closmann (1966), a model was proposed for sulfur deposition in porous media, demonstrating rapid sulfur accumulation near the wellbore during pressure decline. Experimental studies (Shedid et al. 2002) have shown that the intensity of sulfur deposition in carbonate reservoirs depends on the initial concentration, flow rate, and rock permeability. Research has established that low-permeability carbonate formations exhibit significantly higher sulfur deposition rates than high-permeability formations.

In recent years, various analytical and numerical models have been proposed to evaluate the effects of sulfur deposition on reservoir flow characteristics and well inflow performance (Roberts 1997; Al-Awadhy et al. 1998). However, these models typically either do not incorporate sulfur phase transitions under reservoir conditions or offer only simplified descriptions of wellbore deposition dynamics.

This study is based on practical experience from developing a green gas field in the Surkhandarya region of the Republic of Uzbekistan, where persistent sulfur deposition occurs in both reservoir and wellbore systems, creating significant production challenges. It should be noted that this issue exhibits regional characteristics and has not been comprehensively studied through numerical modeling in this area.

The objective of this study is to systematize the accumulated field experience and integrate it into numerical and flow simulation models that account for elemental sulfur phase transitions. To achieve this objective, the following tasks were undertaken:

- construct a reservoir model incorporating sulfur precipitation and its impact on well productivity;
- develop a numerical model of sulfur deposition processes on production tubing walls;
- analyze the influence of key operational factors, including flow rate, tubing diameter, surface operating constraints, and temperature conditions;
- propose a gas flow control strategy that prevents sulfur deposition by ensuring sulfur particle entrainment in the gas flow.

The results of this study enable optimization of well operating parameters, minimize sulfur plugging risks, and enhance field development efficiency in reservoirs containing elemental sulfur in the reservoir fluid.

Geological and Physicochemical Conditions of the Field

Geological Characteristics of the Afghan-Tajik Basin

The geological structure of the field is complex, located in a mountainous area within a tectonically active zone characterized by an extensive deep fault system, some of which extend to the surface. The productive carbonate reservoirs exhibit dual-porosity characteristics: the majority of pore volume is concentrated in the low-permeability matrix, while fluid flow to wells occurs predominantly through natural fracture networks (Gavrilov et al. 2024). Fracture permeability exceeds matrix permeability by several orders of magnitude, resulting in fracture-dominated flow and significant heterogeneity in flow behavior. This complex reservoir behavior has been confirmed by inter-well hydrodynamic interaction data obtained through specialized pressure pulse testing (Gavrilov et al. 2023).

Jurassic Carbonate Reservoir Features

The subsalt carbonate sequence exhibits significant heterogeneity of reservoir properties both laterally and vertically. The reservoir can be subdivided into two distinct zones. The upper zone consists primarily of alternating anhydrites and limestones, whereas the lower zone is composed predominantly of massive, fractured limestones and dolostones (Togaev et al. 2024).

A notable field observation is the absence of sulfur deposits on tubing walls in wells producing from the upper zone. This phenomenon may be attributed to absent or negligible elemental sulfur concentrations in the gas, significant undersaturation with respect to sulfur, or possible sulfur adsorption onto reservoir rock.

Notably, reservoir temperatures in the upper zone range between 115-120°C, which is close to or below the sulfur crystallization temperature. This may promote solid-phase precipitation; however, no deposition is observed in the absence of sulfur in the gas composition. In contrast, the lower zone exhibits higher temperatures of 120-140°C, indicating that elemental sulfur remains in liquid form within the reservoir. This allows the liquid sulfur to participate in fluid flow and be transported with the produced gas.

Reservoir Fluid Description

The reservoir gas composition comprises methane (81.9 mol.%), hydrogen sulfide (6.6 mol.%), carbon dioxide (8.7 mol.%), nitrogen (2.6 mol.%), and minor quantities of higher molecular weight hydrocarbons (Gavrilov et al. 2025). The reservoir gas exists under high pressure and temperature conditions, with reservoir pressure reaching 700 bar and temperature ranging from 115 to 140°C. Based on these thermobaric conditions and composition, the gas is classified as dry. Direct evidence of elemental sulfur presence in wellhead samples is absent. However, it should be noted that no dedicated sulfur detection studies have been performed.

The primary geochemical mechanism responsible for high H₂S concentrations in these deep reservoirs is attributed to thermochemical sulfate reduction (TSR) (Descubes et al. 2021). This process occurs at elevated temperatures (typically >120°C) and involves reactions between sulfate minerals, particularly anhydrite, and hydrocarbons, producing H₂S and CO₂, and potentially elemental sulfur in the gas phase.

Sulfur Phase Behavior in Reservoir and Wellbore Systems

Sulfur Precipitation Mechanisms

Sulfur condensation and crystallization represent the primary mechanisms by which elemental sulfur transitions to the solid phase, occurring both within the reservoir and in wellhead equipment. Under reservoir pressure and temperature conditions, sulfur exists in the gaseous phase. However, as reservoir fluid flow through the reservoir and up the wellbore, decreasing pressure and temperature cause the gas to exceed the sulfur solubility limit, leading to subsequent condensation into the liquid phase.

Upon further cooling, liquid sulfur undergoes crystallization, forming solid crystalline modifications (agglomerates), predominantly rhombic or monoclinic sulfur (Fig. 1a). Crystallization can initiate either spontaneously upon reaching thermodynamic equilibrium or heterogeneously on surface imperfections such as tubing wall roughness, corrosion sites, existing impurities, or deposits.

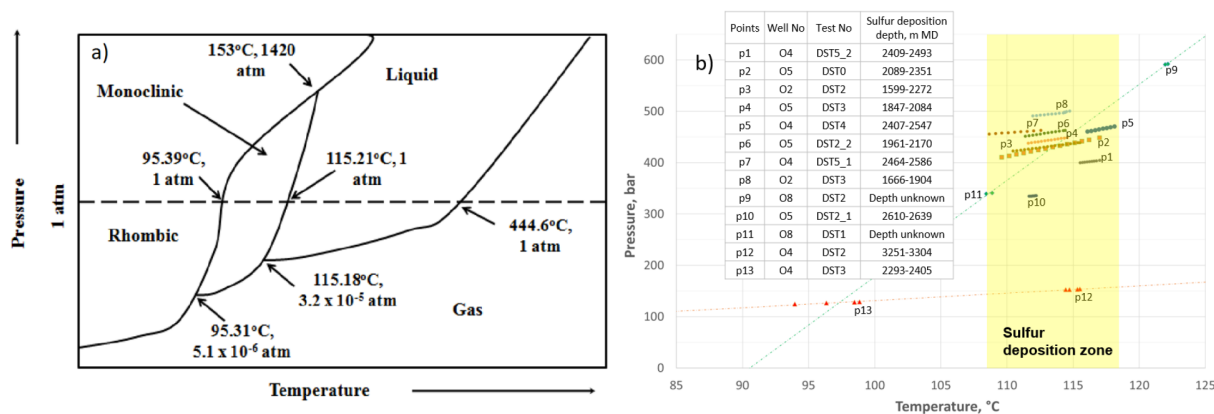


Figure 1—Elemental sulfur phase diagram (Steudel 2003) (a) and P-T conditions in wellbore sections with sulfur deposition zone identified (b)

Solid sulfur deposition progressively accumulates on internal equipment surfaces, diminishing flow capacity and elevating hydraulic resistance. This buildup results in partial flow restriction or, eventually, complete blockage of the flow path, particularly in narrow sections of the tubing string or fittings.

Sulfur Phase Diagram and Its Significance

The sulfur phase diagram provides a basis for predicting the conditions under which phase transitions occur during hydrocarbon production. Elemental sulfur exhibits a complex phase behavior characterized by two distinct solid allotropic forms (rhombic and monoclinic), along with liquid and vapor phases (Fig. 1a).

Following the pressure and temperature trajectory along this diagram during gas flow from the reservoir to the wellhead enables determination of the conditions for sulfur vapor condensation to liquid, followed by crystallization to solid state. The practical significance of the phase diagram lies in its ability to model pressure-temperature profiles where phase transitions occur, which is essential for predicting and preventing sulfur deposition in wellbore equipment.

Based on analysis of production data and pressure-temperature characteristics of several wells, the conditions corresponding to sulfur deposition on tubing inner walls were established (Fig. 1b). According to the phase diagram, elemental sulfur crystallization temperature typically falls within the range of 107–119°C, exhibiting a consistent dependence on pressure: as pressure increases, the crystallization point shifts to higher temperatures. This demonstrates the importance of accurately considering both temperature and pressure when predicting conditions for solid sulfur deposition in the wellbore.

Influence of Pressure and Temperature on Sulfur Phase States

Pressure and temperature are the key parameters determining the phase state of elemental sulfur during gas production. Under the high pressure and temperature conditions typical of deep reservoirs, sulfur remains in gaseous or liquid phase without adhering to wellbore walls or surface equipment. However, as the fluid travels upward through the wellbore, both pressure and temperature decline, creating favorable conditions for phase transitions.

As pressure decreases, dissolved sulfur becomes supersaturated and begins to precipitate into the liquid phase (condenses). Further temperature decline, particularly in the upper wellbore sections and surface equipment, promotes crystallization of liquid sulfur and formation of solid deposits on tubing internal surfaces. The most problematic zones are the wellhead and surface infrastructure components, where temperature drops below the sulfur melting point.

Understanding the influence of pressure-temperature conditions on sulfur phase behavior is of significant practical importance. It enables optimization of well operating regimes by adjusting parameters such as flow rate, wellhead pressure, or heat losses in the tubing string, with the goal of minimizing conditions favorable to sulfur deposition and preventing equipment blockage.

Empirical Model for Sulfur Deposition in Wellbore Tubing

Laboratory Data and Initial Assumptions. Laboratory determination of elemental sulfur solubility in gas is a methodologically challenging task, particularly under high-pressure and high-temperature conditions. Therefore, to develop a model for sulfur deposition processes in the wellbore, an empirical relationship describing sulfur solubility as a function of pressure and temperature was employed, based on the works of Brunner (1988) and Chrastil (1982). This relationship is expressed by the following equation:

$$C_s = \rho_g k \cdot \exp\left(\frac{M}{T} + N\right) \quad (1)$$

The gas density ρ_g under given pressure-temperature conditions is determined by:

$$\rho_g = \frac{\gamma_g \cdot M_{air}}{R} \cdot \frac{P}{T \cdot Z} \quad (2)$$

The gas compressibility factor Z was determined based on laboratory measurements conducted for the gas composition of the field. Using these results, an equation of state model was adapted (Gavrilov et al. 2025), which enabled more accurate calculation of gas properties over a wide range of pressure-temperature conditions.

For solubility model calibration and its application in calculations, field data were utilized: measurements of solid sulfur mass deposited on tubing inner walls at various depths, along with corresponding volumes of produced gas. Through correlation of these data, an empirical relationship was established for the actual deposited sulfur content per unit gas volume, which enabled adaptation of the model parameters (Eq. 1) to field conditions.

Multiphase Flow Regimes and P-T Profile Considerations. Accurate prediction of elemental sulfur equilibrium concentration in the produced gas flow requires precise modeling of pressure and temperature profiles along the wellbore. These profiles directly depend on tubing string diameters, flow rate, reservoir pressure, and well depth. The pressure profile was calculated by numerically solving the gas flow momentum equation for vertical tubing, accounting for gravity, friction, and inertial terms (Eq. 3). The temperature profile was computed using a modified analytical expression that considers heat exchange between the gas and surrounding formation (Eqs. 4–6).

$$\frac{dP}{dz} = -\rho_g \cdot g \cdot \sin\beta - \lambda \cdot \frac{\rho_g \cdot v^2}{2 \cdot D} - \frac{d(\rho_g \cdot v^2)}{2 \cdot dz} \quad (3)$$

$$T_z = T_r - \theta \cdot (L - z) - \Delta T \cdot \exp(-\alpha \cdot (L - z)) + \left(\theta - D_T \cdot \frac{P_{wb} - P_z}{z} - \frac{A}{c_p} \right) \cdot \frac{1 - \exp(-\alpha \cdot (L - z))}{\alpha} \quad (4)$$

$$\Delta T = T_r - T_{wb} = D_T \cdot (P_r - P_{wb}) \cdot \frac{\ln(1 + G \cdot c_g \cdot \tau / (\pi \cdot h \cdot c_r \cdot r_w^2))}{\ln(r_e / r_w)} \quad (5)$$

$$\alpha = \frac{2 \cdot \pi \cdot \lambda_r}{G \cdot c_g \cdot \ln(1 + \sqrt{\pi \cdot \lambda_r \cdot \tau / (c_r \cdot r_w^2)})} \quad (6)$$

To model the dynamics of sulfur deposit accumulation in the wellbore, a system of differential equations describing sulfur transport and deposition in the gas flow was solved using numerical methods. The model core is based on the advection-diffusion equation with sulfur deposition, formulated for sulfur concentration as shown in Eq. 7. The coefficient K depends on supersaturation conditions and is calculated according to the expression in Eq. 8. Changes in sulfur deposit thickness on on tubing inner walls, along with the corresponding reduction in internal diameter, are determined using Eq. 9.

$$\frac{\partial C}{\partial t} + v \cdot \frac{\partial C}{\partial z} = D_s \cdot \frac{\partial^2 C}{\partial z^2} - K \cdot v \cdot (C - C_s) \quad (7)$$

$$K = \begin{cases} 0 & C \leq C_s \\ k_s & C > C_s \end{cases} \quad (8)$$

$$\frac{dD}{dt} = - \frac{2 \cdot K \cdot Q_g \cdot (C - C_s)}{\pi \cdot D \cdot \rho_s} \quad (9)$$

The coefficients λ , k_s , D_s , along with the heat transfer parameters, were calibrated based on actual field data and results from hydrodynamic well studies at the field.

Model Calibration Using Field Data. To ensure proper functioning of the solubility model describing sulfur deposition along the entire gas flow path — from the productive formation to the surface takeoff point — empirical parameters must be calibrated. Calibration was performed using field experimental data obtained during short-term production periods conducted in well testing. Knowing the sulfur mass M_s deposited on tubing inner surfaces and the gas volume N_g produced during that period, it becomes possible to determine the average amount of sulfur precipitated from the gas phase per unit volume of produced gas.

From the model's perspective, this value is defined as the difference between the equilibrium sulfur concentrations at the wellbore bottomhole ($C_{s,wb}$) and at the wellhead ($C_{s,wh}$), and is described by the following expression:

$$\frac{M_s}{N_g} = C_{s,wb} - C_{s,wh} = \rho_{g,wb} k \cdot \exp\left(\frac{M}{T_{wb}} + N\right) - \rho_{g,wh} k \cdot \exp\left(\frac{M}{T_{wh}} + N\right) \quad (10)$$

Since the right-hand side of Eq. 10 is a nonlinear function with respect to the coefficients k , M , and N , a numerical optimization procedure was used to determine them. Specifically, an evolutionary algorithm was applied. The optimization objective was to minimize the root mean square error between calculated and observed values of sulfur amount per unit gas volume. The initial data used for calibration, along with the resulting empirical coefficients, are presented in Table 1

Table 1—Sulfur deposition model calibration results based on field experimental data

Well No	Test No	Bottomhole Data			Wellhead Data			Precipitated Sulfur, g/m ³	Fitted Parameters		
		Pressure, bar	Temperature, °C	Gas Density, kg/m ³	Pressure, bar	Temperature, °C	Gas Density, kg/m ³		k	M	N
E2	DST 3	552	135	280	427	95	269	0.628	3.28	-3593	-9.75
O4	DST 4	541	127	282	360	80	253	0.470	2.62	-3169	-7.32
O4	DST 5	546	126	285	252	73	200	0.415	2.67	-3635	-6.76
O4	DST 6	483	126	264	224	73	180	0.579	2.22	-3553	-3.90
O5	DST 0	485	127	265	295	77	223	0.247	2.64	-3146	-8.12
O5	DST 1	342	123	209	196	57	174	0.345	2.42	-2409	-7.71
O5	DST 2	366	127	218	233	63	196	0.415	2.66	-5300	-1.89
O5	DST 3	501	133	265	337	83	239	0.468	2.97	-4473	-6.16
O5	DST 4	486	129	263	367	84	253	0.291	2.19	-2643	-6.39
O1	DST 0	468	129	257	276	75	213	0.482	2.39	-3650	-4.71
E1	DST 2	449	130	249	312	74	235	0.366	2.62	-3169	-7.32
O10	DST 4	612	139	295	410	62	294	0.463	2.34	-2176	-8.47
O10	DST 5	542	137	276	390	85	263	0.254	3.26	-6376	-4.04
								Average:	2.64	-3638	-6.35

Analysis of Flow Regimes and Their Impact on Deposition. During well testing conducted at the field, continuous monitoring of wellhead and bottomhole pressures, along with temperatures, was carried out. Based on the data obtained, characteristic trends were identified that indicate the influence of hydrodynamic conditions in the wellbore on sulfur deposit formation.

Sulfur deposition begins after a certain period following well startup, after which oscillatory processes are observed in the system. It has been established that wellhead and bottomhole pressures fluctuate in antiphase: during short-term reduction of the effective tubing cross-sectional area, flow rate decreases, bottomhole pressure rises, and wellhead pressure simultaneously drops (Fig. 2a). Similar fluctuations are also observed in temperature characteristics.

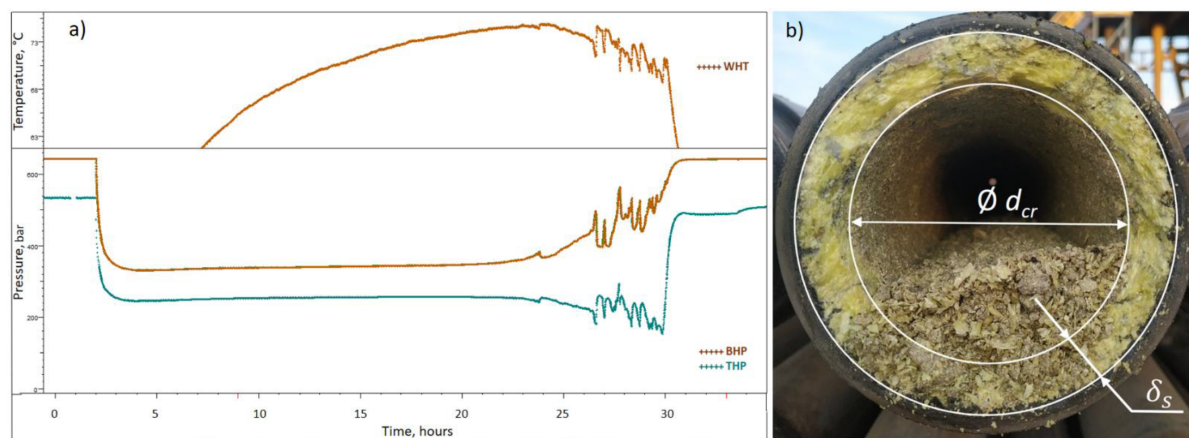


Figure 2—Pressure and temperature oscillations due to sulfur deposition (a) and characteristic sulfur layer formation on the inner tubing wall (b)

After completion of testing and retrieval of the tubing string, sulfur deposits of significant length and specific diameter were found, indicating stable sulfur precipitation on tubing inner walls up to a certain thickness (Fig. 2b). This observation suggests that deposits form until a certain critical flow velocity is reached: when this velocity is exceeded, further sulfur accumulation ceases due to erosional removal of particles by the gas flow.

A minimum critical gas flow rate is required to ensure continuous removal of liquid sulfur from the wellbore bottom. If this condition is not met, liquid sulfur accumulates in the lower wellbore section and is carried upward in slugs, which can lead to rapid blockage of narrow sections and potentially result in production shutdown.

Sulfur solubility modeling has shown that higher wellhead pressure during production (and correspondingly lower drawdown — both attributed to higher productivity wells) reduces sulfur precipitation in the wellbore. A critical flow velocity has also been identified, above which sulfur deposition on walls is prevented — potentially eliminating the issue entirely in some cases. However, excessively high flow rates can lead to accelerated tubing wear due to erosive effects, which limits the practical applicability of such operating modes.

To assess the range of acceptable flow velocities, an erosion equation was applied, calibrated using empirical coefficients derived from field data analysis.

Based on the results obtained, the authors developed a numerical model for sulfur deposition in the wellbore. The model enables forecasting deposit dynamics depending on key operational parameters, determining time intervals of intense deposition, optimizing well operating modes, and taking timely measures to enhance production efficiency while minimizing operational costs.

Modeling Sulfur Impact on Well Productivity

This section describes an integrated approach to modeling the impact of sulfur deposits on well productivity using a multiphase reservoir simulation model. The main focus is on developing and applying a three-phase model that accounts for sulfur solubility, changes in relative permeabilities, and local modeling of sulfur accumulation zones in the near-wellbore zone. The results obtained enable quantitative assessment of productivity losses and support justification of optimal well operating constraints.

Three-Phase Simulation Model (Gas – Liquid Sulfur – Water)

Within the framework of this study, a three-phase reservoir simulation model was developed, including the following fluid phases: gas, liquid sulfur, and water. In the model, precipitated sulfur is treated as a separate "oil" phase. The distribution of elemental sulfur between the gas and "oil" phases is defined by the "condensate"/gas ratio. The influence of sulfur content on other gas phase properties necessary for modeling (such as volumetric expansion coefficient and viscosity) was considered negligible. The model accounts for sulfur phase transitions that occur when reservoir pressure drops below the saturation pressure, leading to sulfur precipitation from the gas phase into the liquid phase. The reservoir temperature is assumed constant (isothermal approximation), which corresponds to conditions typical for most deep reservoirs.

The physicochemical properties of liquid sulfur (high viscosity, low fluidity, and tendency to aggregate) were accounted for by introducing appropriate equations of state and corrections to phase permeability. Special attention in the model is given to liquid sulfur saturation in the near-wellbore zone and modeling its accumulation in the pore space (the "sulfur banking" effect), which critically impacts the reservoir's flow characteristics.

The core of the model consists of material balance equations for each phase (Eq. 11) and the generalized Darcy equation (Eq. 12)

$$\frac{\partial}{\partial t}(\phi \cdot S_i \cdot \rho_i) + \nabla(\rho_i \cdot \mathbf{v}_i) = q_i \quad i = \{g, s, w\} \quad (11)$$

$$\mathbf{v}_i = -\frac{k \cdot k_{ri}}{\mu_i} \cdot \nabla P_i \quad (12)$$

The developed model served as the basis for calculating the impact of sulfur deposits on relative permeabilities, well productivity, and the effectiveness of various production enhancement methods.

Sulfur Solubility and Relative Permeability Functions

An empirically calibrated model (Eq. 1) was used as the basis for describing elemental sulfur solubility in natural gas, enabling quantitative estimation of elemental sulfur content in the gas phase depending on pressure and temperature. According to this model, the initial elemental sulfur concentration in the gas does not exceed 1.74 g/m³ under the given reservoir conditions. These parameters are used to calculate the onset of sulfur precipitation and for subsequent modeling of its accumulation in pore space.

The degree of flow impairment caused by deposited sulfur, in addition to pore volume reduction and pressure drop — which reduce sulfur solubility — is also determined by gas phase relative permeability. This permeability, in turn, depends on the spatial distribution of the deposited phase. In this study, relative phase permeability relationships for the gas-liquid sulfur system (Fig. 3a) were based on laboratory core studies (Guo et al. 2022). For the gas-water system (Fig. 3b), relationships obtained from laboratory core analysis conducted on actual reservoir samples from the field were applied.

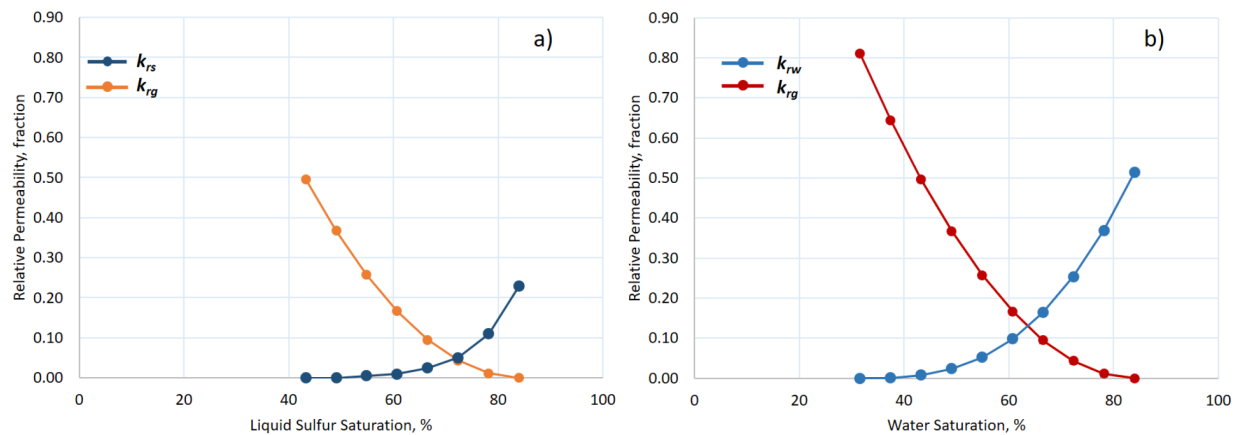


Figure 3—Relative permeability curves for gas–liquid sulfur (a) and gas–water (b) systems

Analysis of the relative permeability curves for the gas-liquid sulfur system shows that the critical saturation at which liquid sulfur begins to participate in flow is 43%. The zone of simultaneous flow of gas and liquid sulfur is narrow, corresponding to a liquid sulfur saturation range from 43% to 83%. Beyond this saturation range, a sharp decline in gas phase relative permeability is observed.

Sulfur Banking Simulation Using Local Grid Refinement

Since the major pressure drop during gas flow occurs in the near-wellbore zone, local grid refinement (LGR) was used in model construction. This approach allowed for more accurate description of liquid sulfur accumulation processes near the well bottom, known as the sulfur banking effect – the formation of a zone with increased sulfur saturation that leads to reduction of gas phase permeability.

The use of a refined grid in the near-wellbore region (Fig. 4) provided high spatial resolution for the model, which is critically important for analyzing phase distribution and deposit dynamics within reservoir void space. Based on numerical modeling, an assessment was conducted of the temporal and spatial evolution of sulfur deposits both in the reservoir and near-wellbore zone. The calculations showed that the highest risk of blockage and sharp permeability decline occurs within a few meters of the wellbore, where a stable zone of liquid sulfur accumulation forms.

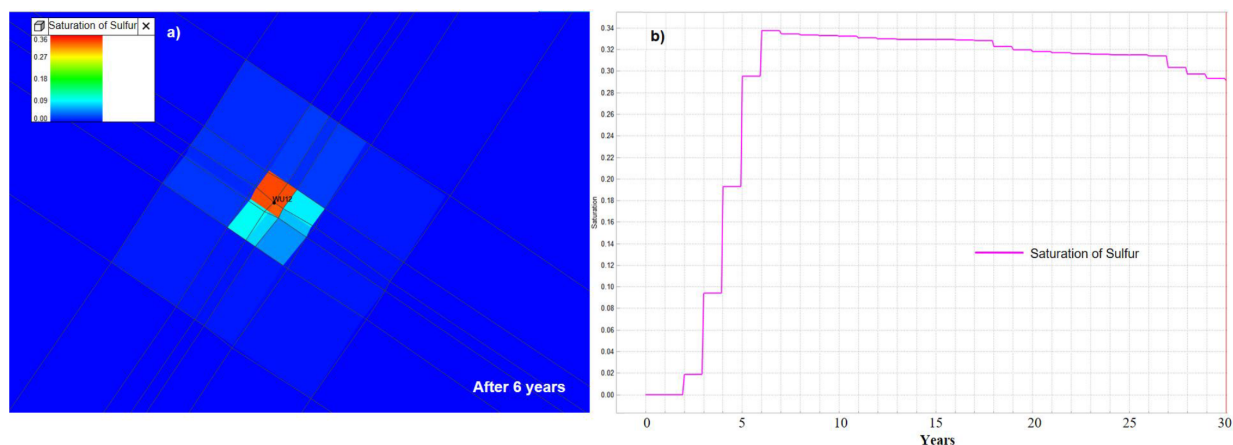


Figure 4—Dynamic of sulfur saturation changes: (a) in the near-wellbore zone after 6 years; (b) in the near-well cell over a 30-year period

The results obtained confirm the necessity of more detailed near-wellbore region modeling when designing well operating constraints and selecting reservoir stimulation methods to prevent or mitigate sulfur deposition.

Sulfur Deposition Impact on Well Productivity and Operating Constraints

Using the developed model, an analysis of sulfur deposit impact on well productivity was performed. A series of numerical experiments demonstrated productivity decline depending on the rate of sulfur precipitation and changes in gas phase relative permeability. The analysis showed that the higher the drawdown, the earlier productivity decline occurs due to sulfur deposition in the reservoir.

The actual impact of well productivity on sulfur deposit intensity in the tubing was also analyzed by comparing production results at different reservoir drawdown levels (Fig. 5). The results showed that tubing blockage caused by sulfur deposition occurred significantly earlier at higher drawdown values. To ensure longer and more stable well operation, an optimal drawdown limit of no more than 30 bar was established. This regime corresponds to the "green zone" in Fig. 5, within which the likelihood of intensive sulfur deposition is minimal.

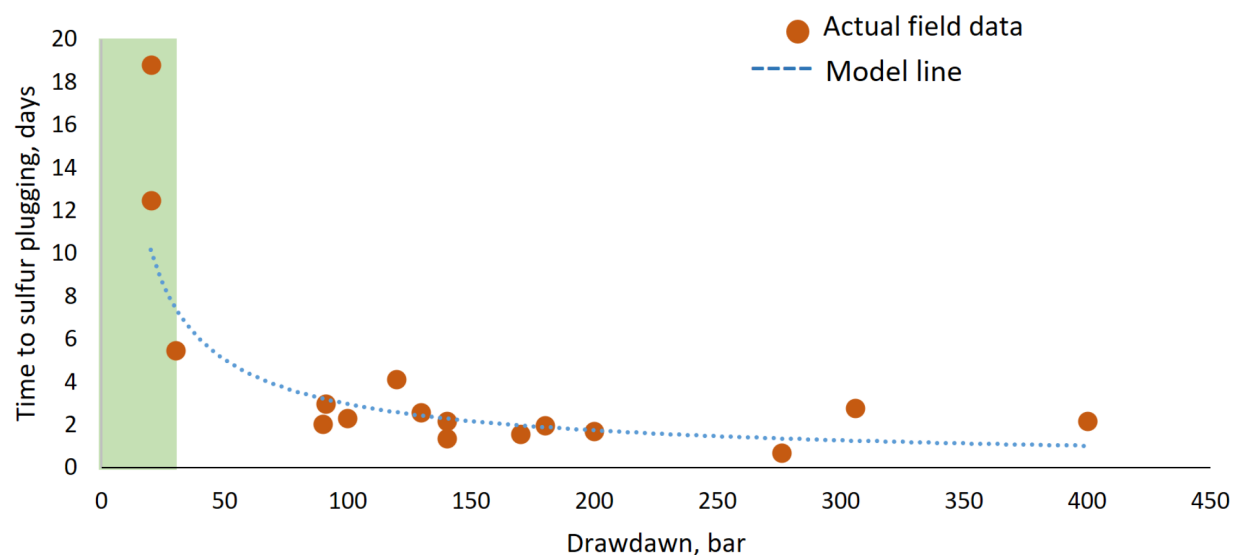


Figure 5—Well production time before tubing plugging vs. reservoir drawdown

Results and Discussion

The solubility equation for elemental sulfur in sour gas (Chrastil 1982) was calibrated based on actual sulfur deposition data in tubing obtained during well testing at the field. The results of the calibrated solubility model for various temperatures are presented in Fig. 6a. The graph shows that at constant temperature, pressure decrease from 650 to 500 bar leads to sulfur solubility reduction of approximately 50%, corresponding to precipitation of half the sulfur mass from the gas phase as deposits. Further pressure reduction to 200 bar results in over 95% of sulfur precipitating, confirming the high intensity of sulfur deposition at large pressure drops. Temperature decrease from 140°C to 100°C at constant pressure also causes sulfur solubility to drop by about half, leading to up to 50% sulfur precipitation from the gas phase.

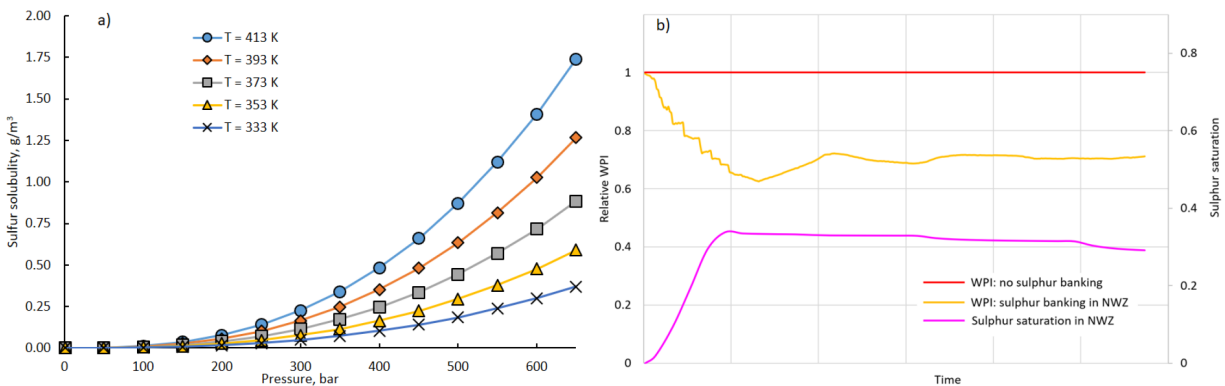


Figure 6—Calibrated solubility model of elemental sulfur in natural gas as a function of pressure and temperature (a) and simulated well productivity index decline due to sulfur banking in the near-wellbore zone (b)

Simulation results (Fig. 6b) showed that the well productivity index (WPI) decreases by up to 30% during the first year of production due to liquid sulfur accumulation in the near-wellbore zone. This effect is caused by gas phase permeability reduction in the reservoir due to sulfur precipitation, the intensity of which strongly depends on the magnitude of reservoir drawdown. The data obtained indicate significant impact of sulfur deposits in the reservoir on well productivity, especially during early development stages. Therefore, this factor must be considered when designing field development and selecting well operating constraints.

With a constant well operating configuration (including wellhead choke diameter, tubing size, and IPR – inflow performance relationship), a gradual increase in sulfur deposit thickness on tubing inner walls is observed over time. As shown in Fig. 7, sulfur deposit formation begins at a certain depth and develops in a stepwise manner. Each subsequent "step" adds to the deposit thickness over the previous layer, progressively narrowing the tubing cross-sectional area. Upon reaching critical narrowing of the cross-sectional area (Fig. 2b), gas flow velocity increases to a level at which erosive removal of sulfur agglomerates is observed. Under such conditions, further sulfur deposition may occur either higher up the wellbore closer to the wellhead, or lower down towards the bottomhole – depending on the temperature profile relative to the crystallization temperature of elemental sulfur. This leads to extension of the deposition zone and increasing sulfur deposit thickness along the tubing inner surface where temperature is below the elemental sulfur crystallization temperature. These processes are accompanied by production rate decrease due to reduced tubing flow capacity. Upon reaching critical sulfur deposit thickness, sudden sulfur plug formation may occur as a portion of liquid sulfur arrives and instantly solidifies. This phenomenon is accompanied by sharp reduction in wellbore flow capacity and may result in complete gas production stoppage. Fig. 7 also presents pressure and temperature profiles along the wellbore. The graphs show that as sulfur deposit length and thickness increase, pressure and temperature below the deposition zone tend to rise, while above the zone they tend to decrease.

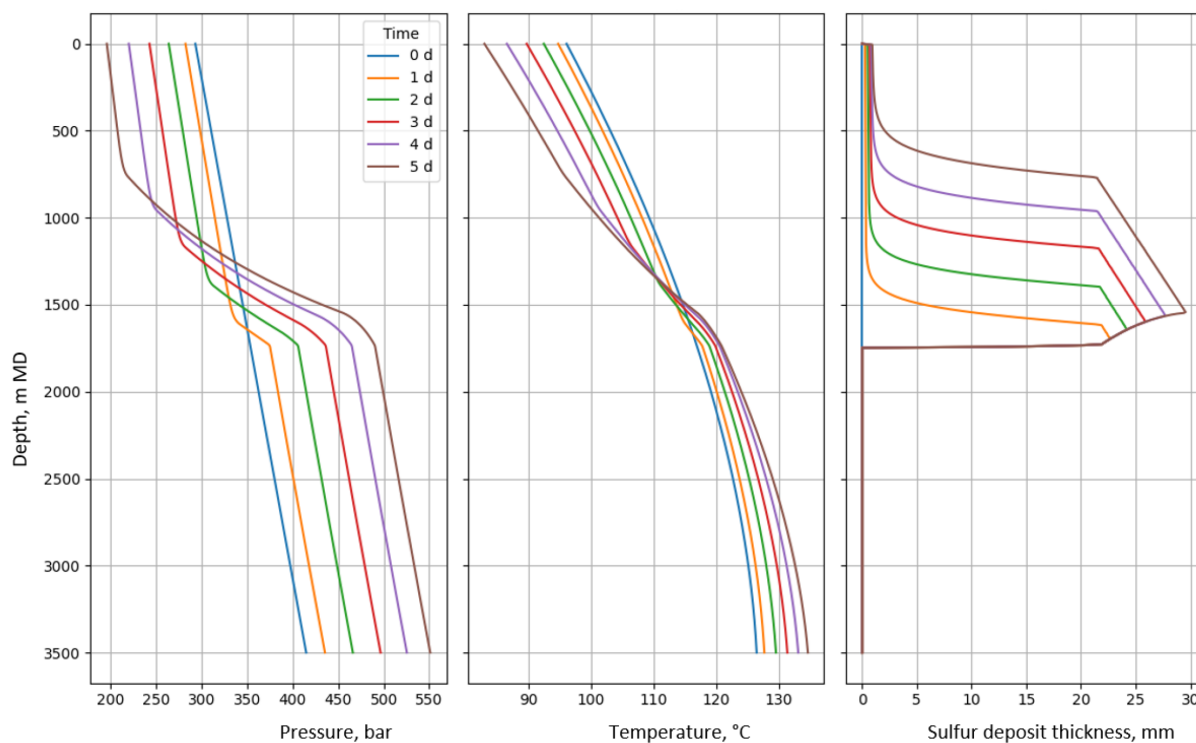


Figure 7—Simulated pressure, temperature, and sulfur deposit thickness profiles along the wellbore at different production times

Within the sulfur formation interval, a characteristic overlap zone forms where a mixed trend is observed. Furthermore, this active sulfur deposition region features distinct curve inflections, reflecting abrupt changes in hydrodynamic conditions within the well.

To assess the impact of well production rate on sulfur deposition processes in the wellbore over a fixed production period, the wellhead choke diameter was varied to half and double the baseline value. As shown in Fig. 8, increasing gas flow rate by enlarging the choke diameter leads to decreased pressure profile along wellbore. The pressure reduction, in turn, decreases sulfur solubility in the flow, promoting increased total mass of deposited sulfur. At the same time, gas flow velocity increases, which affects sulfur deposit morphology – resulting in decreased maximum deposit thickness. The onset of sulfur deposition shifts toward the bottomhole due to increased reservoir drawdown and, consequently, lower temperature in the near-wellbore zone. Moreover, once gas flow velocity reaches critical sulfur transport velocity, further sulfur accumulation in the wellbore ceases. Thus, to minimize sulfur deposits, it is recommended to operate well at flow rates exceeding this critical velocity. It should also be noted that while sulfur deposition zone length increases with production rate, maximum deposit thickness decreases.

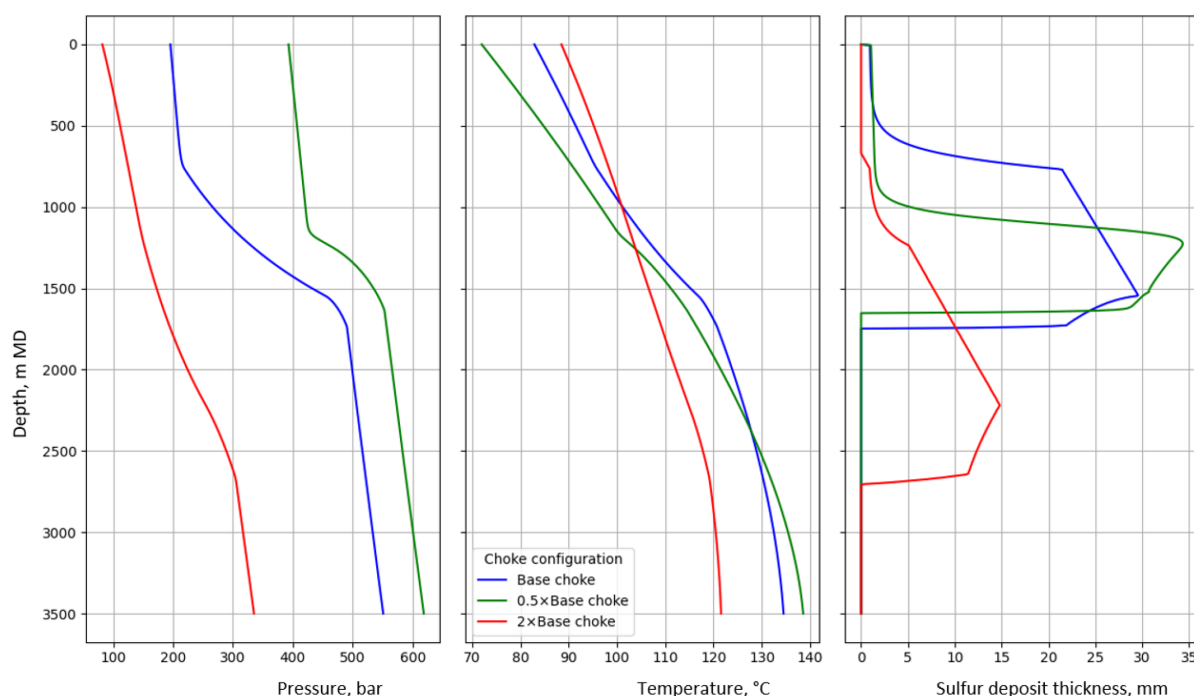


Figure 8—Influence of wellhead choke diameter on pressure and temperature profiles, deposition zone interval, and sulfur deposit thickness in the wellbore

To determine the influence of reservoir productivity on the dynamics of sulfur deposit formation, calculations were performed by varying the parameters of the well IPR model (C and n). Parameter C was varied by factors of 0.5 and 2.0 relative to the base case, correspondingly halving and doubling the well's AOF rate. At the same time, wellhead choke diameter, tubing type, and production time remained unchanged. According to results presented in Fig. 9, with increasing well productivity, pressure and temperature profiles along most of the wellbore increase, sulfur solubility rises, and sulfur deposition onset shifts closer to the wellhead. At fixed production time, increased maximum thickness, total mass, and length of sulfur deposits is observed with rising well productivity. However, the upward shift of the deposition zone indicates existence of a critical productivity threshold, above which the sulfur deposition interval extends beyond the wellbore. This means that conditions favorable for sulfur precipitation within the wellbore do not develop. This effect is typically associated with the wellhead temperature exceeding the elemental sulfur crystallization temperature. Thus, by selecting optimal thermal operating constraints in highly productive wells, continuous production without sulfur deposits in the wellbore can be ensured.

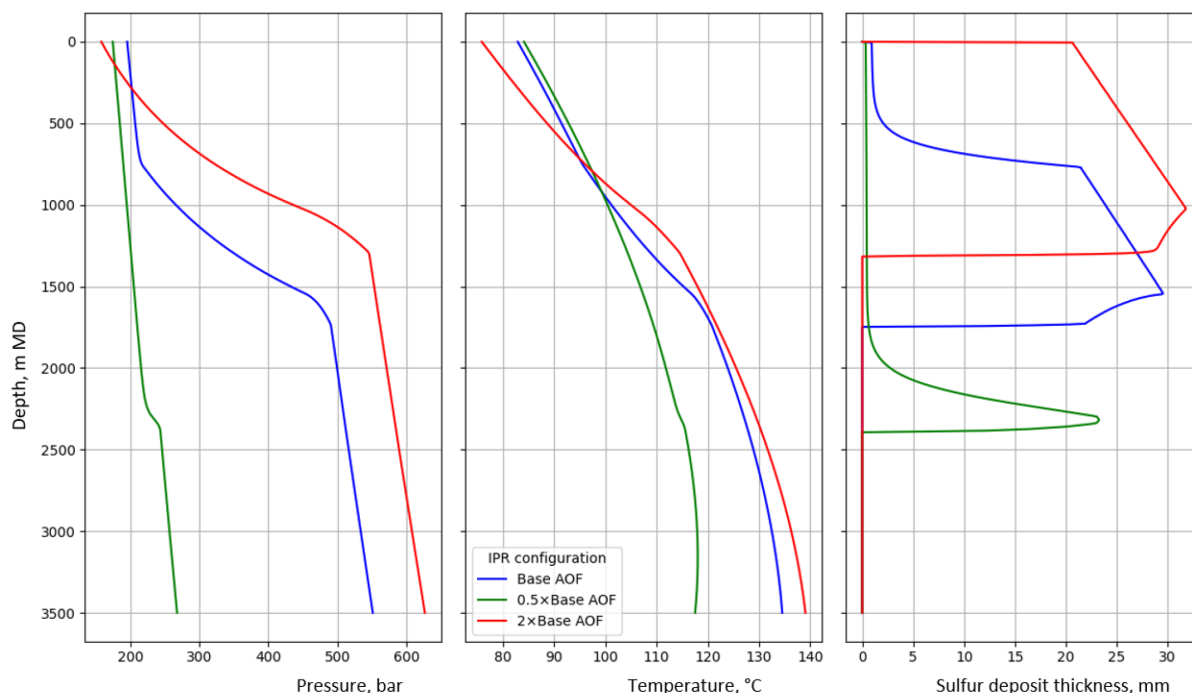


Figure 9—Effect of well productivity on pressure, temperature, and sulfur deposition profiles in the wellbore

The next stage of analysis involved studying the impact of tubing internal diameter on sulfur deposit dynamics in the wellbore, while keeping reservoir parameters, wellhead choke diameter, and production duration constant. The model considered three tubing diameter options: the base diameter, half the base diameter, and twice the base diameter. Calculation results are presented in Fig. 10. According to data obtained, increasing tubing internal diameter leads to higher flow rates and longer periods of stable well production without signs of blockage. Decreasing internal diameter results in increased pressure and temperature gradients along the wellbore – that is, sharper pressure and temperature drops from bottomhole to wellhead. After sulfur deposits form, this effect intensifies: the pressure and temperature difference between bottomhole and wellhead becomes even more pronounced. This is due to additional flow area narrowing caused by deposits, which increases hydraulic resistance and reduces heat transfer. As a result, there is increasing pressure and temperature drop along the wellbore, especially at small tubing diameters. When tubing diameter is increased, the sulfur deposition onset zone shifts closer to the wellhead, with maximum deposit thickness increasing and deposition interval length decreasing. Thus, increasing tubing inner diameter can be an effective method to mitigate sulfur deposits: upon reaching a certain diameter, complete sulfur transport beyond the wellhead is possible, preventing accumulation within the wellbore.

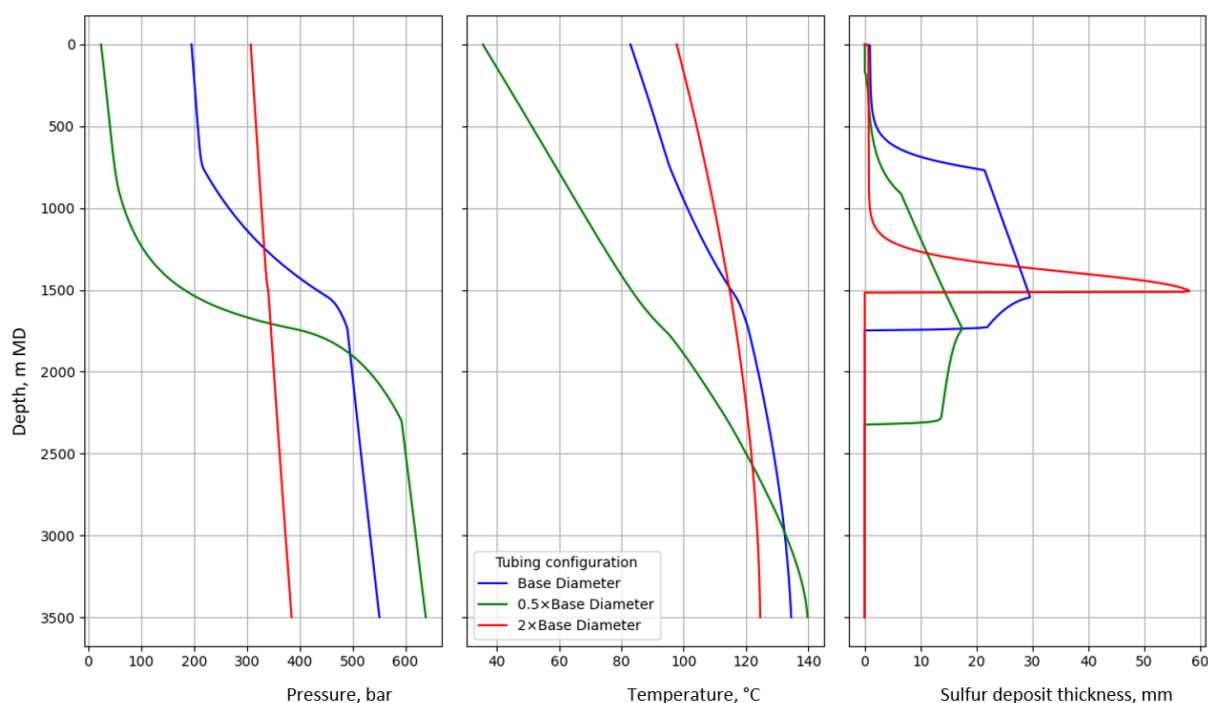


Figure 10—Impact of tubing inner diameter on pressure, temperature, and sulfur deposition profiles along the wellbore

Conclusions

In this study, a numerical approach was developed and implemented for modeling elemental sulfur precipitation and deposition processes both in the wellbore and productive reservoir during gas production in sour gas field. The results obtained enabled the following key conclusions:

1. The developed sulfur solubility model, based on experimental data and a modified Chrastil equation, demonstrates a strong sensitivity of sulfur solubility to pressure and temperature variations. A decrease in these parameters leads to a sharp reduction in solubility, resulting in intensive sulfur precipitation from the gas phase. Numerical simulation results indicate that during the first year of well operation, productivity may decline by up to 30% due to the formation of elemental sulfur in the near-wellbore zone, which significantly reduces gas permeability. These effects must be accounted for when establishing well operating parameters and developing field strategies.
2. Sulfur deposition in the wellbore is a cumulative process that progressively reduces the flow cross-sectional area. Once a critical threshold is reached, rapid formation of sulfur plugs may occur, potentially causing complete blockage of the flow path and production cessation. Analysis of field welltest data enabled the identification of an optimal operational envelope, specifically maintaining drawdown pressure below 30 bar, under which the risks of intense sulfur deposition and plug formation are minimized, thereby ensuring extended periods of stable well performance without the need for mechanical cleanouts.
3. Increasing well flow rate by increasing wellhead choke diameter leads to greater deposited sulfur mass; however, increased gas flow velocity limits the maximum deposit thickness and shifts the deposition zone closer to the bottomhole. Once critical sulfur transport velocity is reached, further sulfur accumulation in the wellbore ceases.
4. Increasing well productivity leads to higher sulfur solubility in the wellbore and shifts the deposition zone upward along the wellbore. At a certain productivity level, temperature and pressure conditions are established that practically prevent sulfur deposition in the well, enabling prospects for uninterrupted production with optimal selection of operating constraints.

5. Tubing diameter significantly influences sulfur deposition dynamics. Increasing tubing diameter reduces flow resistance, decreases heat loss, and can ensure complete sulfur transport beyond the wellhead, making this approach a promising method for preventing sulfur plugging.

Overall, the proposed methodology and simulation model developed by the authors enable prediction of elemental sulfur behavior within the reservoir-wellbore system and can be effectively used to optimize gas well operation in sour gas fields facing sulfur deposition issues. The results obtained are recommended for use at the studied field in designing field development and production facilities, justifying operating constraints, and choosing engineering solutions aimed at preventing sulfur deposits and plugging.

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Nomenclature

C_s	= Instantaneous sulfur solubility, kg/m ³
ρ_g	= Natural gas density, kg/m ³
T	= Temperature, K
P	= Pressure, Pa
Z	= Gas compressibility factor (Z-factor)
k, M, N	= Empirical coefficients in the sulfur solubility equation
γ_g	= Gas specific gravity
M_{air}	= Molar mass of dry air, 28.96 kg/kmol
R	= Gas constant, 0.08315 bar·m ³ /(kmol·K)
g	= Gravitational acceleration, m/s ²
β	= Wellbore inclination angle, deg
λ	= Friction factor
v	= Gas flow velocity, m/s
D	= Wellbore diameter, m
T_r	= Formation temperature, K
θ	= Geothermal gradient, K/m
α	= Thermal diffusivity of wellbore to formation, m ² /s
L	= Wellbore bottomhole depth, m
z	= Depth from wellhead, m
D_T	= Joule-Thomson coefficient, K/Pa
r, wb, wh	= Subscripts for reservoir, bottomhole, and wellhead conditions
G	= Mass flow rate of gas, kg/s
c_g	= Specific heat capacity of gas, J/(kg·K)
c_p	= Specific heat capacity of formation rock, J/(kg·K)
t, τ	= Production time
h	= Pay zone thickness, m
r_w	= Wellbore radius, m
λ_r	= Formation thermal conductivity, W/(m·K)
C	= Sulfur solubility in gas, kg/m ³
D_s	= Sulfur effective mass diffusion coefficient, m ² /s
K, k_s	= Sulfur mass transfer coefficient, 1/m

- Q_g = Gas production rate, m³/s
 ρ_s = Sulfur density, kg/m³
 M_s = Mass of deposited sulfur, kg
 δ_s = Sulfur deposit thickness on wellbore wall, m
 d_{cr} = Critical flow diameter, m
 N_g = Cumulative gas production, m³
 ϕ = Formation porosity
 $i=\{g,s,w\}$ = Gas, sulfur, and water phases
 S_i = Phase saturation
 k = Absolute permeability, mD
 k_{ri} = Relative permeability to phase *i*
 μ_i = Viscosity of phase *i*, cP

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